

Sakshi Tapkir

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Summary

Software engineer with experience building REST APIs and backend systems using Java, Spring Boot, Node.js, and Python, including recent work on LLM and RAG systems. Skilled in data modeling, secure API design (JWT, RBAC), and automated testing and CI/CD, delivering production-ready backend features on Agile teams and owning projects through implementation.

Education

Northeastern University

Boston, MA

Master of Science in Computer Software Engineering

Aug. 2024 – Aug 2026

Coursework - Database Management & Design, Software Quality Control, Agile, Computational Skepticism, Web Development Tools

Savitribai Phule Pune University

Pune, MH

Bachelor in Computer Science

Aug. 2020 – Jun 2023

Coursework - Data Structures & Algorithms, Distributed Systems, Systems Design, Digital Networking, Operating Systems

Experience

Software Engineer Intern

Feb 2026 – Apr 2026

Humanitarians AI

Boston, MA

- Built a multi-model AI platform (React, Node.js, Express, Supabase) gathering requirements from 100+ faculty researchers, consolidating six separate LLM-powered assistants into one interface, eliminating the need to switch between tools per research session.
- Engineered a parallel LLM execution engine with Node.js and Promise.all that issues six concurrent assistant API calls, cutting panel-mode response time from 18s to 3s and making real-time side-by-side comparison practical.
- Orchestrated a RESTful Express routing layer with JWT authentication that normalized six inconsistent assistant API response formats into one predictable schema, removing the need for format-specific error handling on the frontend.
- Containerized the application with Docker and automated CI/CD through GitHub Actions, adding Supabase row-level security and request-level tracing to support a weekly production release cadence with fewer manual deployment steps.

Software Development Engineer

Jul 2023 – Jun 2024

MR SON Equipments Pvt Ltd

Pune, MH

- Designed an inventory REST API with Spring Boot and MySQL, modeling the schema for real-time stock tracking, barcode lookup, and automated low-stock alerts that replaced manual spreadsheet checks and flagged shortages before they delayed orders.
- Developed a customer order-tracking module with Spring Boot, Hibernate ORM, and Redis, weighing indexed database lookups against a cache layer to cut order-retrieval time by 40% on a dataset of 500+ orders.
- Secured 10+ ERP microservices with an authorization layer in Spring Security, combining JWT authentication, role-based access control for three roles, request validation, and per-endpoint rate limiting to block unauthorized access.
- Automated unit and integration tests for critical service paths and maintained technical documentation, working in an Agile/Scrum process with Git to catch regressions before release and keep delivery predictable.

Projects

Eventopia: Event Platform | Next.js, React.js, Node.js, MongoDB, Express.js, Stripe, Clerk, REST APIs | [GitHub](#)

- Architected a full-stack event marketplace using Next.js, React, and MongoDB with ticket checkout, organizer event management, and an idempotent Stripe payment flow using webhook verification and unique transaction IDs to prevent duplicate charges.
- Optimized the event-listing path by parallelizing queries, removing a redundant order aggregation, and adding caching and a covering index, cutting p95 latency from 3.7s to 1.2s and tripling throughput to nearly 99 req/s at 200 concurrent users under k6 load testing.

Document Parser | Python, FastAPI, Pydantic, LangChain, FAISS, NumPy, Ollama, SQLite, Supabase | [GitHub](#)

- Engineered a RAG pipeline (Python, FastAPI, LangChain, FAISS) for document-grounded Q&A over PDF/TXT with hybrid retrieval, reranking, and confidence-gating that decides whether to answer or abstain based on evidence quality.
- Validated answer quality with an automated LLM-as-judge evaluation harness (74 pytest tests, CI), reducing hallucinated answers from 39% to 33% against a no-retrieval baseline across 18 cases.

Certifications

Advanced Certification in Data Science and AI

Feb 2023 – Feb 2024

IIT Madras

India

Technical Skills

Languages: Java, Python, JavaScript, C++, TypeScript, SQL

Backend: Spring Boot, Node.js, Express.js, Hibernate ORM, REST APIs, OAuth, Kafka, GraphQL, WebSockets

Databases: MySQL, PostgreSQL, MongoDB, Supabase, Redis

Cloud & DevOps: AWS (S3, EC2), Azure, Docker, Kubernetes, GitHub Actions, CI/CD, Git, Jira

AI & ML: PyTorch, LangChain, FAISS, RAG, LLM, Supabase Vector Store, Pandas, Scikit-learn

Frontend: React.js, Next.js, Angular, HTML, CSS

Testing: JUnit, Unit Testing, Integration Testing